Pico 2 mini OS Lancer

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Contents

Starting
Playing
The lance
The pod
Enemies
The stages and the bosses
Game over and high scores
Your own sprites
Notes

Lancer is a side-scrolling space shooter. Your ship flies through three stages of caves and fortresses, against waves of enemies, guns on the walls and a boss at the end of each stage. You fire a plain shot, or hold fire to charge the lance, a beam that goes straight through what it kills. Shoot down a hauler and it leaves a capsule. The first coloured capsule brings the pod, which locks on to your ship, shields it, and fires a weapon of its own. You can launch the pod and call it back. The game runs at 60 frames a second on the HDMI screen and plays its sound and music over HDMI. All of its art, sound and music were made for this game.

Starting



Type `lancer` at the shell. The title screen takes turns showing the keys and the five best scores. **Enter** (or **Fire**, or a gamepad's Start) starts a game, and **Esc** goes back to the shell.

Key	What it does
Arrows, or W A S D	move the ship (eight ways)
Space, Z or Ctrl	fire: tap to shoot, hold to charge the lance
X, C or Shift	launch the pod, or call it back
P	pause
M	sound on or off
N	music on or off (the sound effects stay)
Esc	end the game (back to the title); on the title, back to the shell
F10	back to the shell from anywhere

A USB gamepad works too: the stick or D-pad moves the ship, the first and third buttons fire, the second and fourth work the pod, and Start starts a game or pauses it.

Playing



The screen scrolls slowly to the right, and you can fly anywhere on it. Touching a wall, an enemy or an enemy shot loses your ship. The walls are as dangerous as the enemies: some stretches narrow down, step up or down, or have pillars and boulders in the way.

The panel at the bottom shows the beam's charge, your ships in reserve, your speed-ups (green chevrons), **M** when you have missiles, the pod's colour (with a small cross at its second level), your score, the high score and the stage.

You start with three ships. You get an extra ship at 50,000 points and then one every 100,000. When a ship is lost, the next one starts at the last restart point of the stage (there are two in each stage, and one at the boss) without its pod, speed-ups or missiles. A hauler carrying a pod capsule comes along soon after, so you can pick up a pod again.

The lance



Tap fire for the plain shot. Hold it and the beam charges: the meter at the bottom fills and a glow builds at your ship's nose. Let go to fire the lance. A third of the meter gives a small lance, two thirds a bigger one, and a full meter (it flashes) the biggest. A lance goes on through every enemy it kills. The full-size one goes on through enemies it doesn't kill too, and hits each one once. Walls and armour stop it.

The pod



Shoot down a **hauler** (the slow purple carrier) and it leaves a capsule. A **P** capsule changes colour as it drifts: red, blue, yellow. The first one you pick up brings the pod, which flies in and locks on to the nose of your ship in that colour. The pod:

- stops enemy shots that hit it (but not a boss's laser),
- hurts any enemy it touches, again and again,
- fires its colour's weapon each time you tap fire.

Colour	Weapon
Red	a spread of shots ahead (three, or five at level 2)
Blue	lasers on the diagonals that bounce off the walls (and one straight ahead at level 2)
Yellow	shots up and down that then run along the floor and the ceiling (and a plain shot at level 2)

Another P capsule of the same colour raises the pod to level 2 (after that it is worth 1000 points). One of another colour changes the pod's colour.

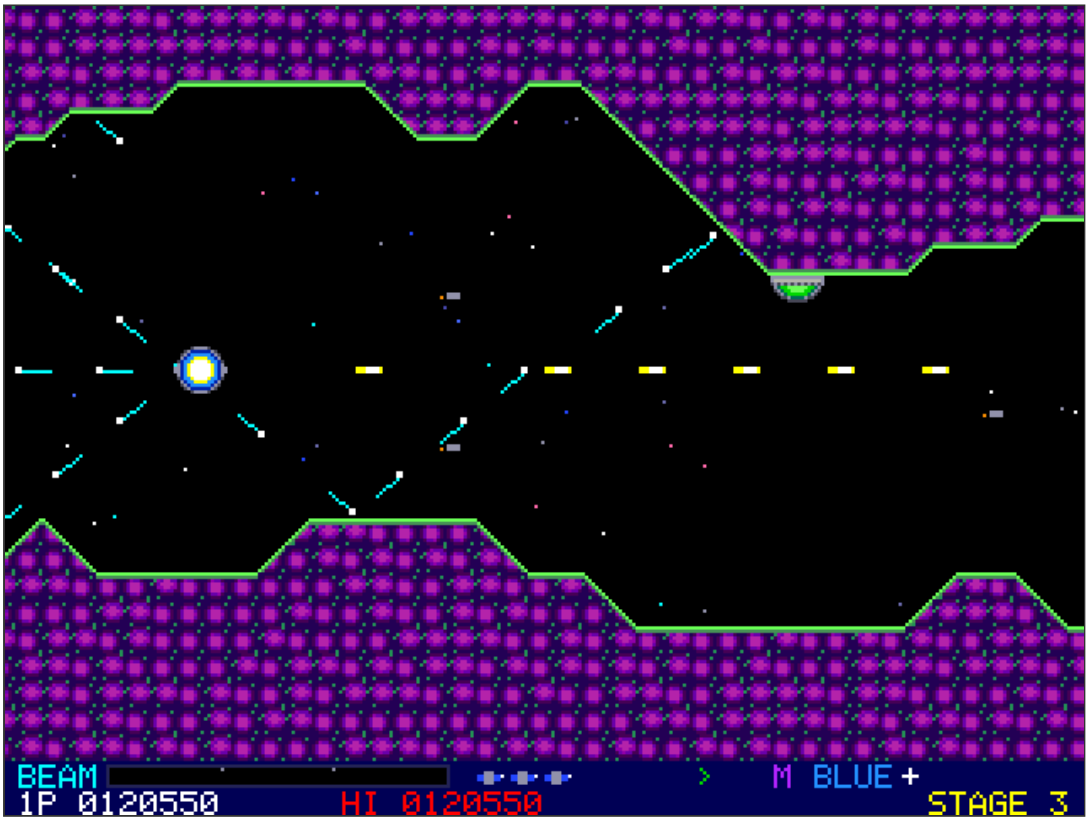


Press **X** (or C, or Shift) to launch the pod. It flies off the way it faces and then stays on its own, keeping level with your ship, firing straight ahead (and behind at level 2) when you fire, and still stopping shots and hurting enemies. Send it into a group of enemies, or park it in front of a gun. Press X again to call it back: it flies to your ship and locks on to the side it reaches. You can also fly into a pod that is on its own to pick it up, on either side. On the back of your ship it fires its weapon backwards, which helps when enemies come from behind.



The other capsules:

Capsule	What it does
S (green)	a faster ship (up to three times)
M (purple)	two homing missiles with each shot (one pair every two thirds of a second)



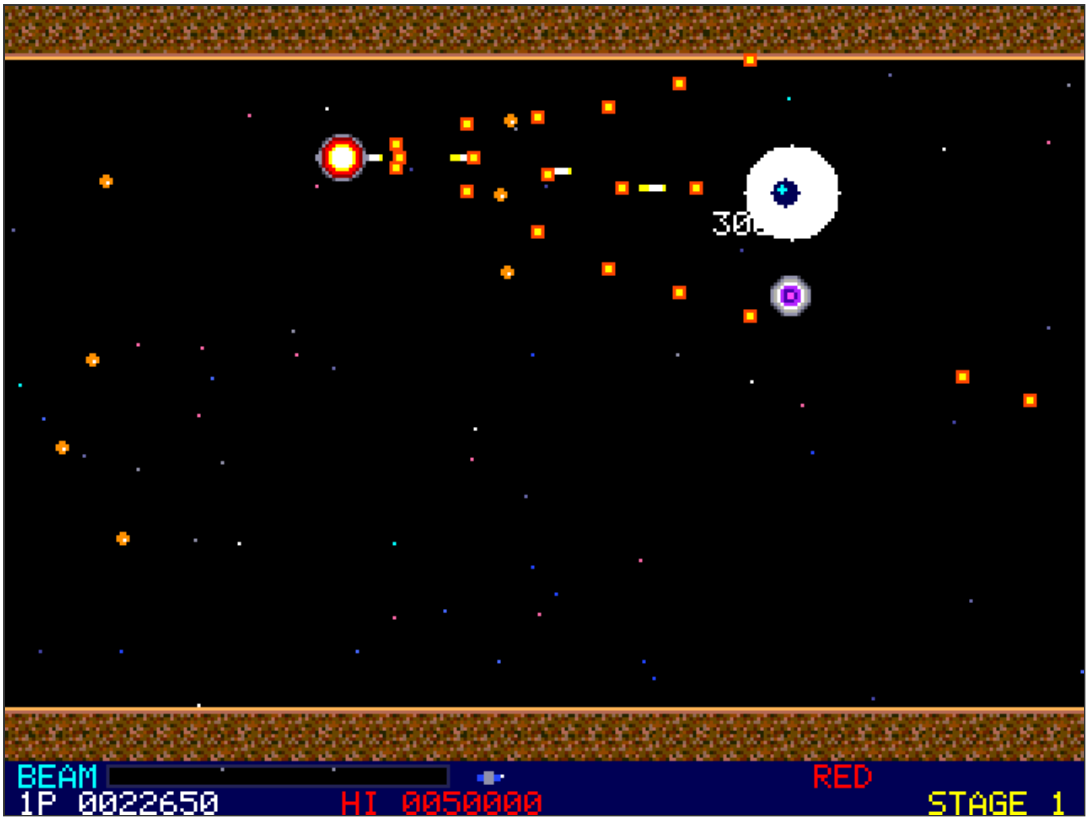
Enemies

Enemy	Points	What it does
Dart (red)	100	flies in waves along a curve; some fire
Swooper (green)	150	flies in and turns at you; some come from behind and turn back
Gun dome	300	sits on the floor or the ceiling and fires at you
Walker	400	walks along the floor or the ceiling and fires from the top of its hops
Hauler	500	slow; leaves a capsule
Mine	200	drifts towards you, and bursts into six shots when it is shot
Coil	200 a ball, 500 the head	a chain of balls following its head along a wave; the head fires
Spore hatch	800	opens now and then to let out spores; armoured while shut
Spore	50	a small seeker

The stages and the bosses



1. **The Rubble Belt**: rock caves with slopes and boulders. Its boss, the **Orrery**, is a great eye ringed by eight armour plates that go round it. Shoot a way through the plates or send the pod in. It fires at you in threes and sends out rings of shots, and when it is badly hurt its plates spin faster and it calls in darts.
2. **The Iron Citadel**: a metal fortress of steps and pillars. Its boss, the **Bastion**, is a fortress block that rides up and down with four guns on its face. Now and then its hatch opens: that is the only time it can be hurt. A red dotted line warns that its laser is about to fire along that row, so get out of the way. With its guns gone, its hatch opens more often and it fires rings.
3. **The Hive**: living walls, narrow and winding. Its boss, the **Wyrm**, winds round the arena. Its body is armour and kills on touch; only its head and the spiked tail can be hurt. The tail fires rings until it is shot off.



A **WARNING** comes up when a boss is near, and the screen stops scrolling for the fight. Beating a boss is worth 10,000 times the stage number. After the third stage the game starts over from the first, as round 2: the enemies fire faster and more often and the bosses take more hits, and so on each round.




BEAM
1P 0152100
HI 0152100
M RED
STAGE 3

Game over and high scores

If your score is one of the five best when your last ship is gone, you enter your initials. Type the three letters, or step through the letters with the arrows and press Fire for each one. Backspace goes back a letter.



The five best scores are kept in `/LANCER.HI` on the USB drive, a short text file (`score NAME` on each line). Delete it to start the table again. If there is no drive, the scores last until you leave the game.


LANCER
- BEST PILOTS -
1 0203100 KEN
2 0050000 ACE
3 0040000 POD
4 0030000 PIX
5 0020000 USB
PUSH ENTER
ESC TO LEAVE

Your own sprites

The game's sprites can be changed with the sprite editor:

```
lancer sprites LANCER.SPR
sprite LANCER.SPR
```

`lancer sprites` writes the built-in sprites as a .SPR file with 21 frames of 16x16. The frames are, in order: the ship (two frames, its flame flickering); the pod (two); the dart (two); the swooper (two); the gun dome; the walker (two steps); the hauler (two); the mine (two); a coil ball and a coil head; the spore hatch shut and open; the capsule; the Orrery's armour plate. The ship faces right and the enemies left. Draw things that sit on the ground standing on the bottom row; the game turns them upside down for the ceiling. Draw the pod and the capsule in red: the game makes the blue, yellow, green and purple ones by changing the reds. When `/LANCER.SPR` (at the top of the drive) is a 16x16 file with at least 21 frames, the game uses it in place of its own art. Rename or delete the file to go back to the built-in art.

Notes

- While the game runs, the screen is 320x240 with each pixel shown as two by two. Each frame is drawn in a second buffer in SRAM and copied to the screen during the vertical sync, so nothing flickers. The stage's map, the pictures and everything else the game uses live in the free part of the framebuffer, so it keeps no memory once you leave.
- The game takes over the sound, so the internet radio stops while you play.
- On a PC, `tools/lancertest/uitest.sh [folder]` builds the game and lets an autopilot play a whole game through all three bosses. It checks the stages' maps, the pod and its weapons, the lance, the bosses and the high-score file, then writes the screenshots in this manual, a picture of each stage's whole map, and every sound effect and tune to `lancer_sounds.wav`.